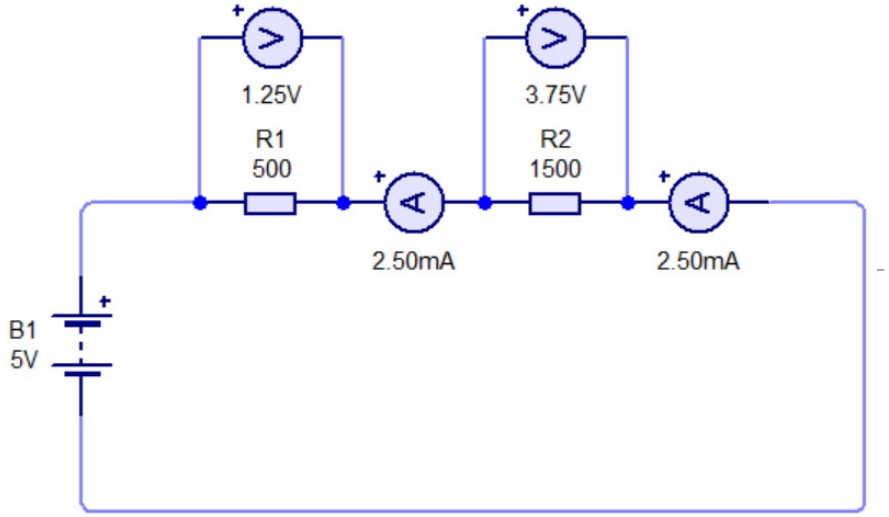
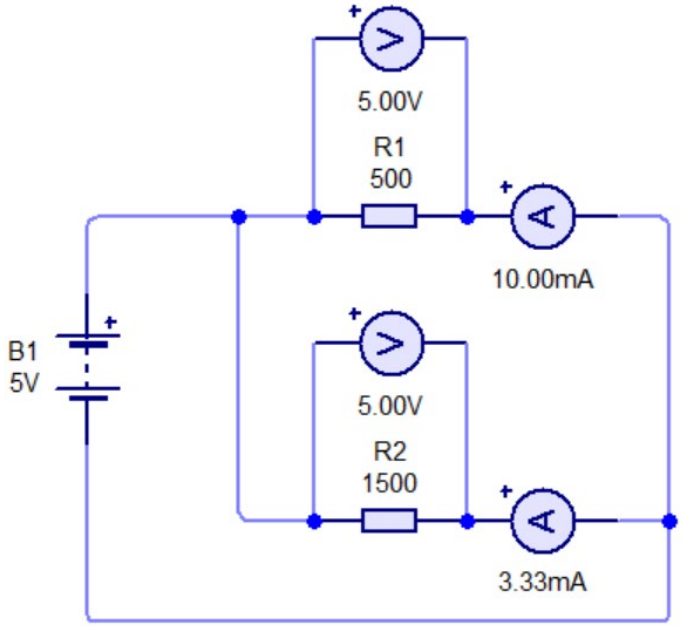
Kirchoff’s Laws



Recap on DC Circuits

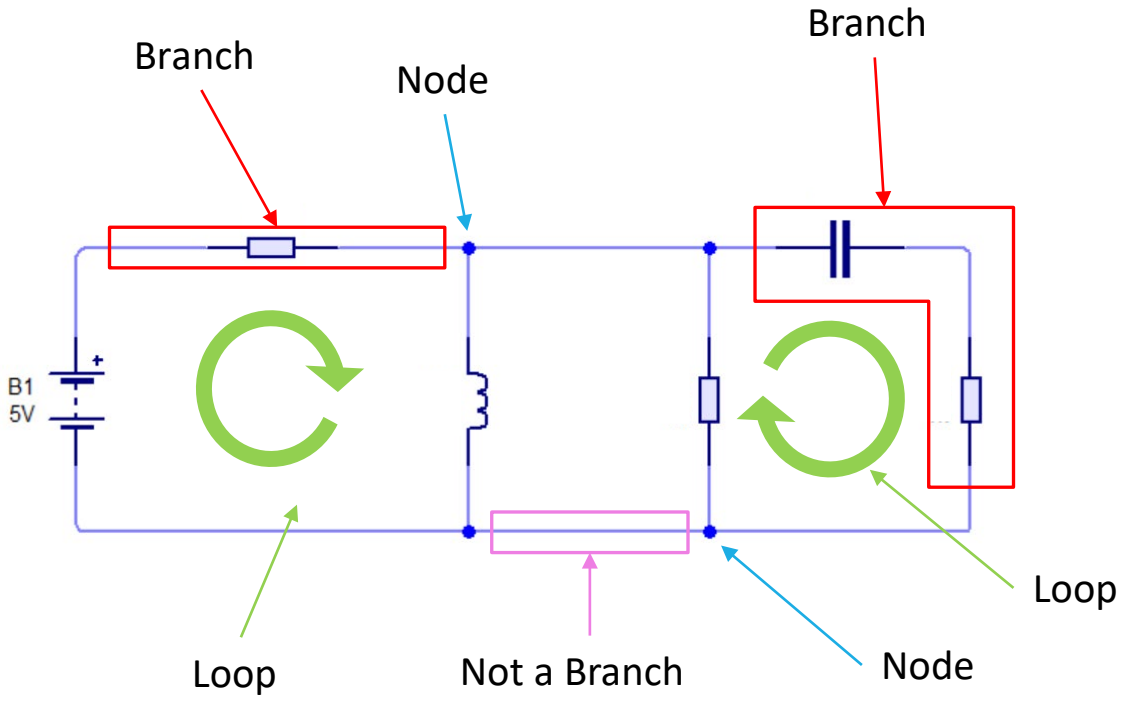
- Ohms law (V=IR)
- **Series connection:** Current is the same through all components, voltages split
- **Parallel connection:** Voltage is the same across branches, currents split





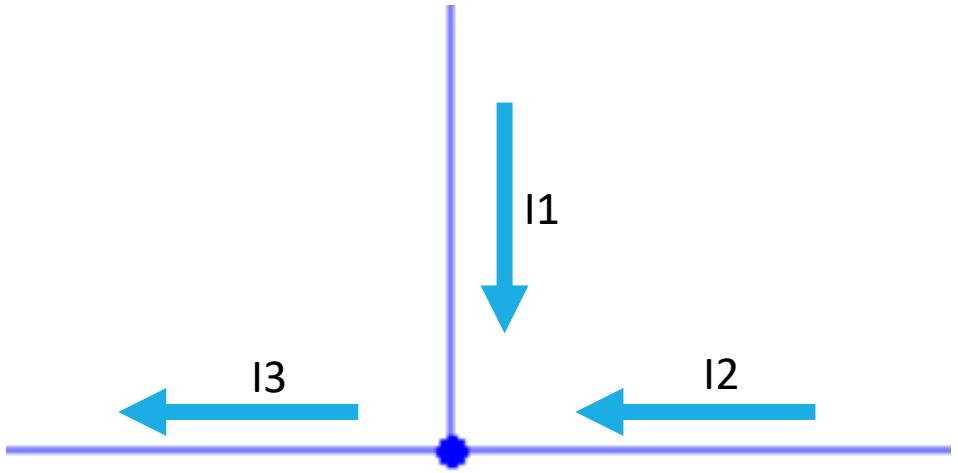
Parts of DC Circuits

- **Node**: A point in a circuit where two or more components are connected.
- **Branch**: A single path connecting two nodes that contains a circuit element (e.g., resistor, voltage source).
- **Loop/Mesh**: Any closed conducting path in a circuit where you can start at one point, travel through elements, and return to the starting point without retracing.



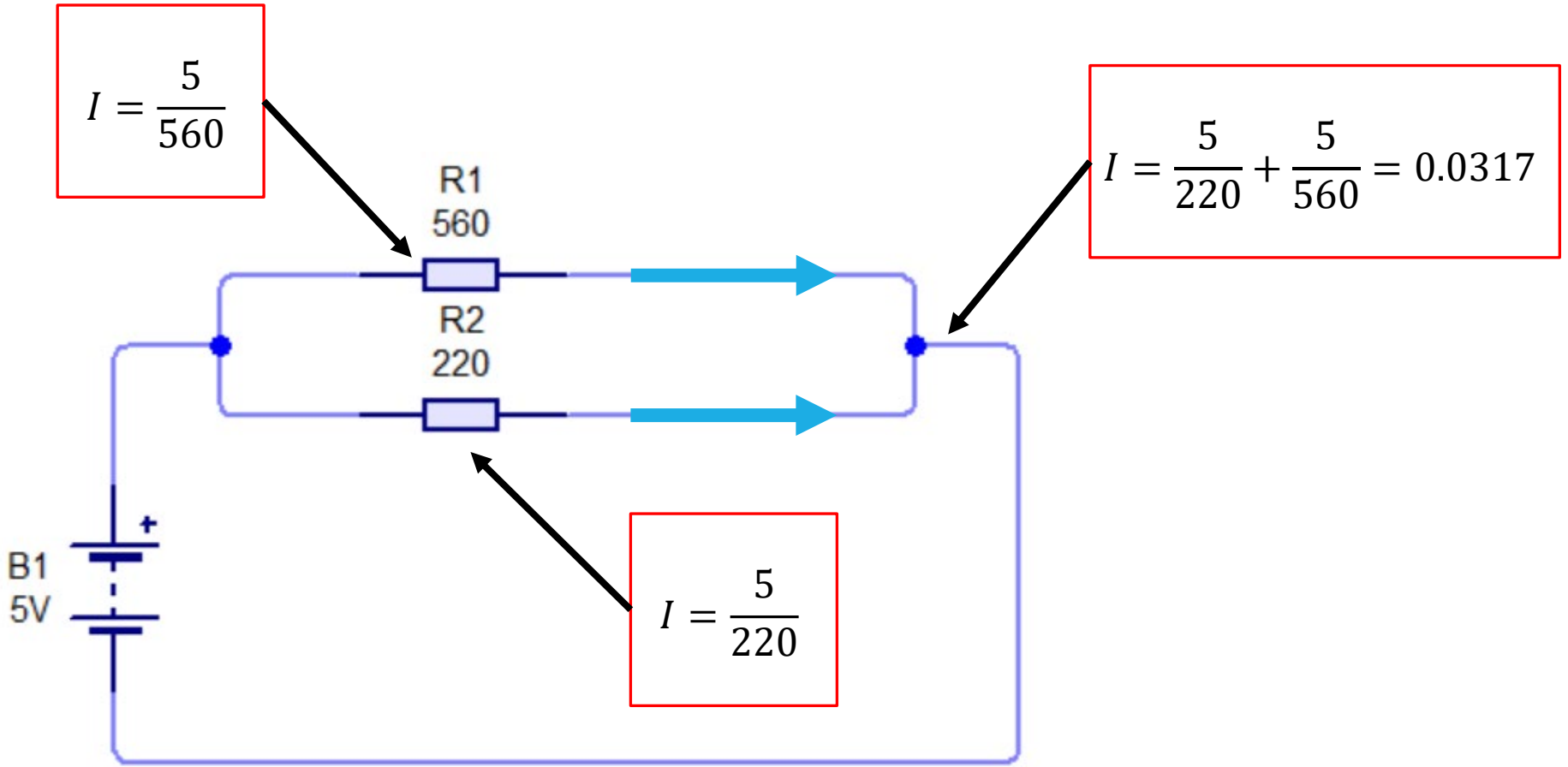
Kirchoff's Current Law

- **Definition:** At any junction, the total current entering = total current leaving
- **Equation form:** $\sum I_{in} = \sum I_{out}$
- **Basis:** Conservation of charge



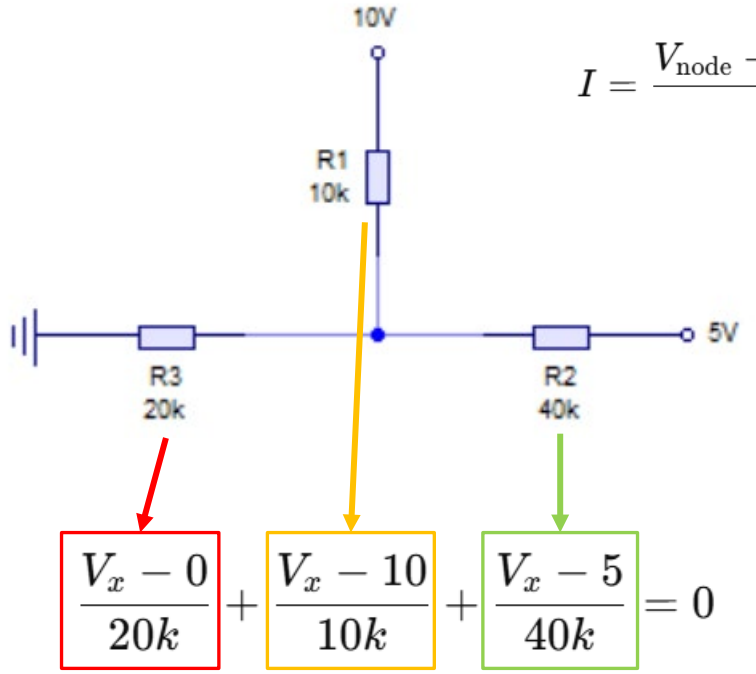
$$I1 + I2 = I3$$

KCL Example


$I = \frac{5}{560}$
$I = \frac{5}{220} + \frac{5}{560} = 0.0317$
R1
560
R2
220
B1
5V
+
$I = \frac{5}{220}$

Example Solving a Kirchoff's Current Question #1

- A typical KCL question will ask you to find the voltage at the node
- We know that $I = \frac{V}{R}$
- We also know $I_1 + I_2 + I_3 = 0$
- So, we can write $\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_2}{R_2} = 0$
- We only need to look at the voltage at the node, so we need to know the voltage drop across the resistors
- We can rearrange it to get the equation below the circuit
- Then we can rearrange to get V_x

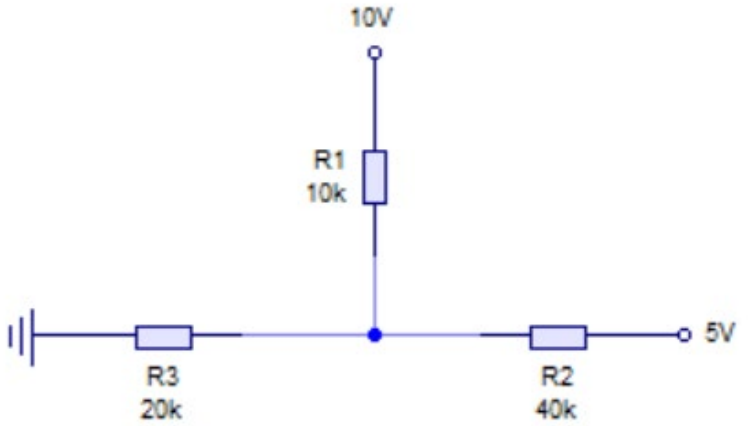


$$I = \frac{V_{\text{node}} - V_{\text{other side}}}{R}$$

$$\frac{V_x - 0}{20k} + \frac{V_x - 10}{10k} + \frac{V_x - 5}{40k} = 0$$

Example Solving a Kirchoff's Current Question #1

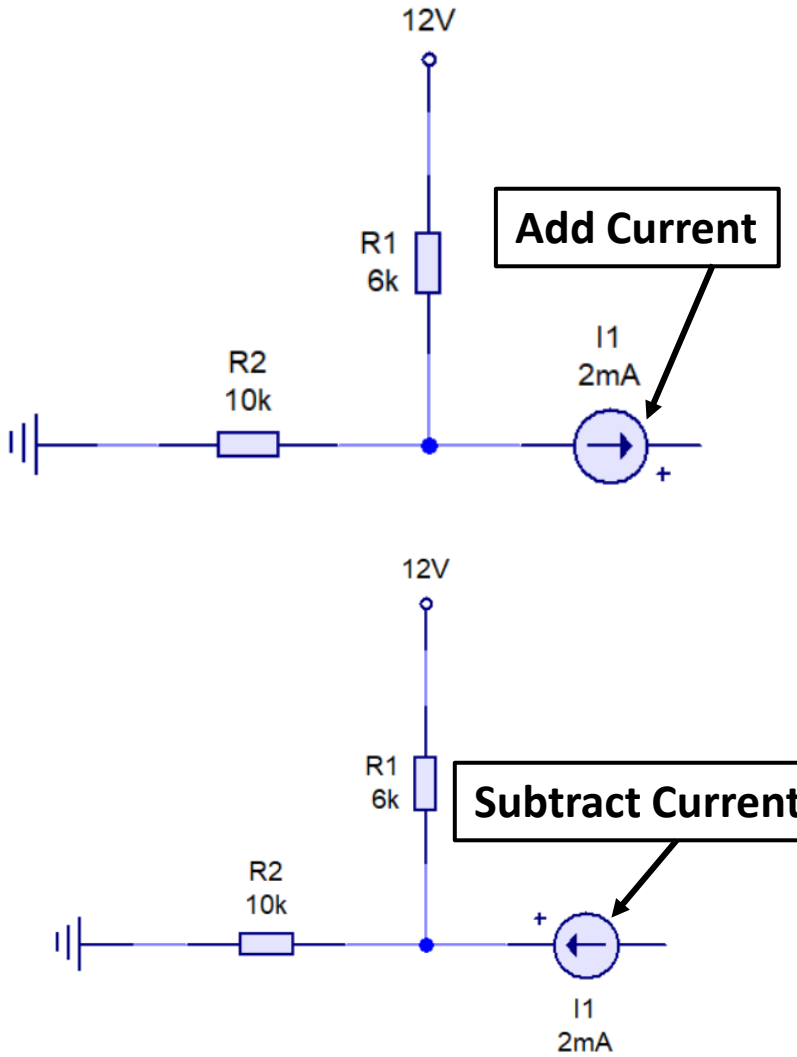
- Then we can rearrange to get V_x
- To get V_x by itself we can get rid of the denominators, to do this we multiply everything by the biggest denominator
- $\frac{40000}{20000}(V_x - 0) + \frac{40000}{10000}(V_x - 10) + \frac{40000}{40000}(V_x - 5)$
- $2(V_x - 0) + 4(V_x - 10) + 1(V_x - 5)$
- $2V_x - 0 + 4V_x - 40 + V_x - 5$
- $7V_x - 45 = 0 \quad \therefore \quad V_x = 6.4286$



$$I = \frac{V_{\text{node}} - V_{\text{other side}}}{R}$$

Current Sources

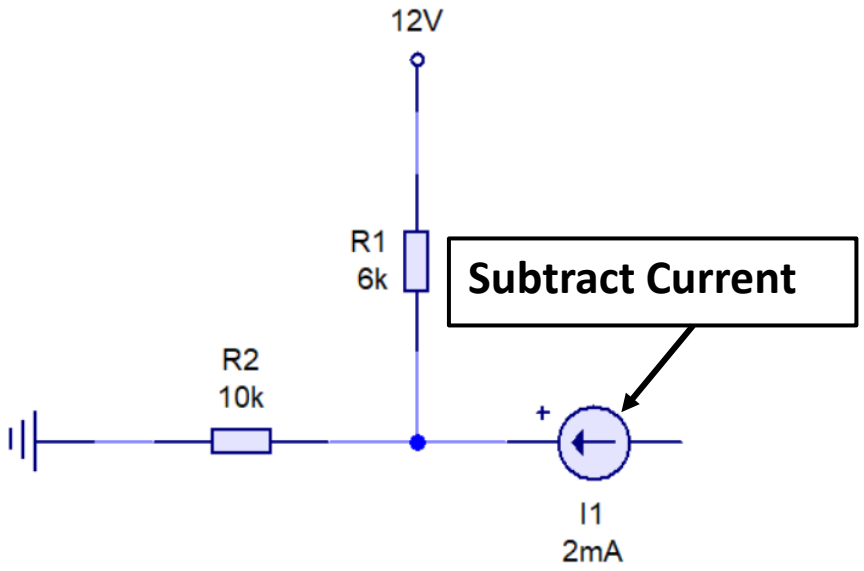
- Some circuits have current sources in them
- They are represented by an arrow in a circle
- The direction of the arrow is important, if the arrow is pointing away current is leaving the node.
- When the arrow is pointing away from the node you add the current
- If the arrow is pointing towards the node its supplying current to the node.
- When the arrow is pointing towards the node you subtract the current



Current Sources

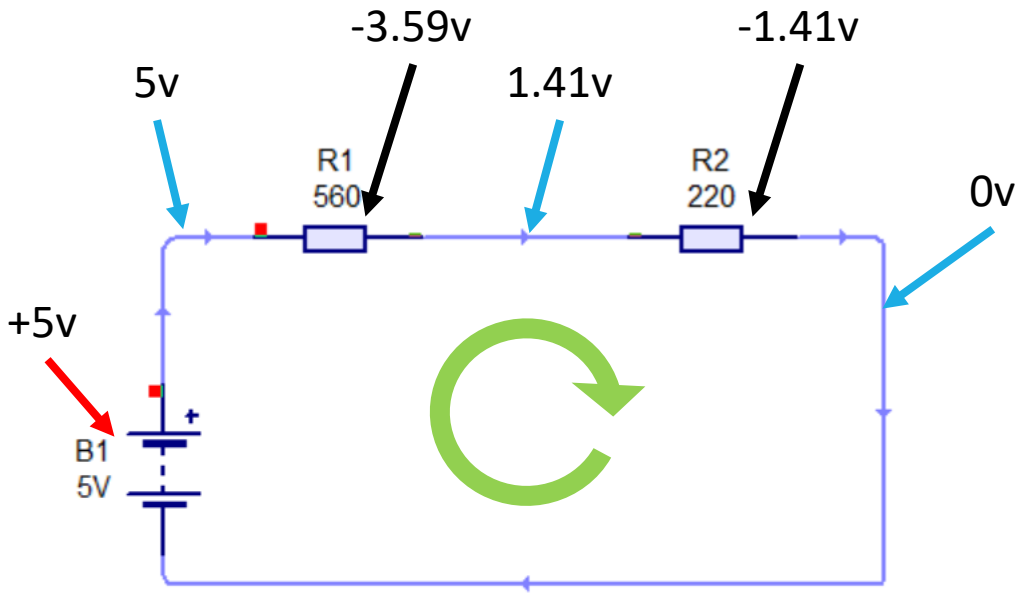
- So if we have the circuit on this slide
- We write out our equation making sure to subtract the current
- $\frac{V_x-12}{6000}+\frac{V_x-0}{10000}-0.002$
- Then we multiply everything by the largest denominator
- $\frac{10000}{6000}(V_x-12)+\frac{10000}{10000}(V_x-0)-(0.002*10000)$
- $1.\dot{6}(V_x-12)+V_x-0-20$

$1.\dot{6}V_x-20+V_x-20\rightarrow 2.\dot{6}V_x-40\rightarrow V_x=24$



Kirchoff's Voltage Law

- **Definition:** The sum of all voltages around any closed loop in a circuit is zero.
- **Equation form:** $\sum V = 0$
- **Meaning:** Energy is conserved—voltage rises (sources) are balanced by voltage drops (loads).
- **Rule of thumb:** When you go around a loop, add rises as positive, drops as negative.

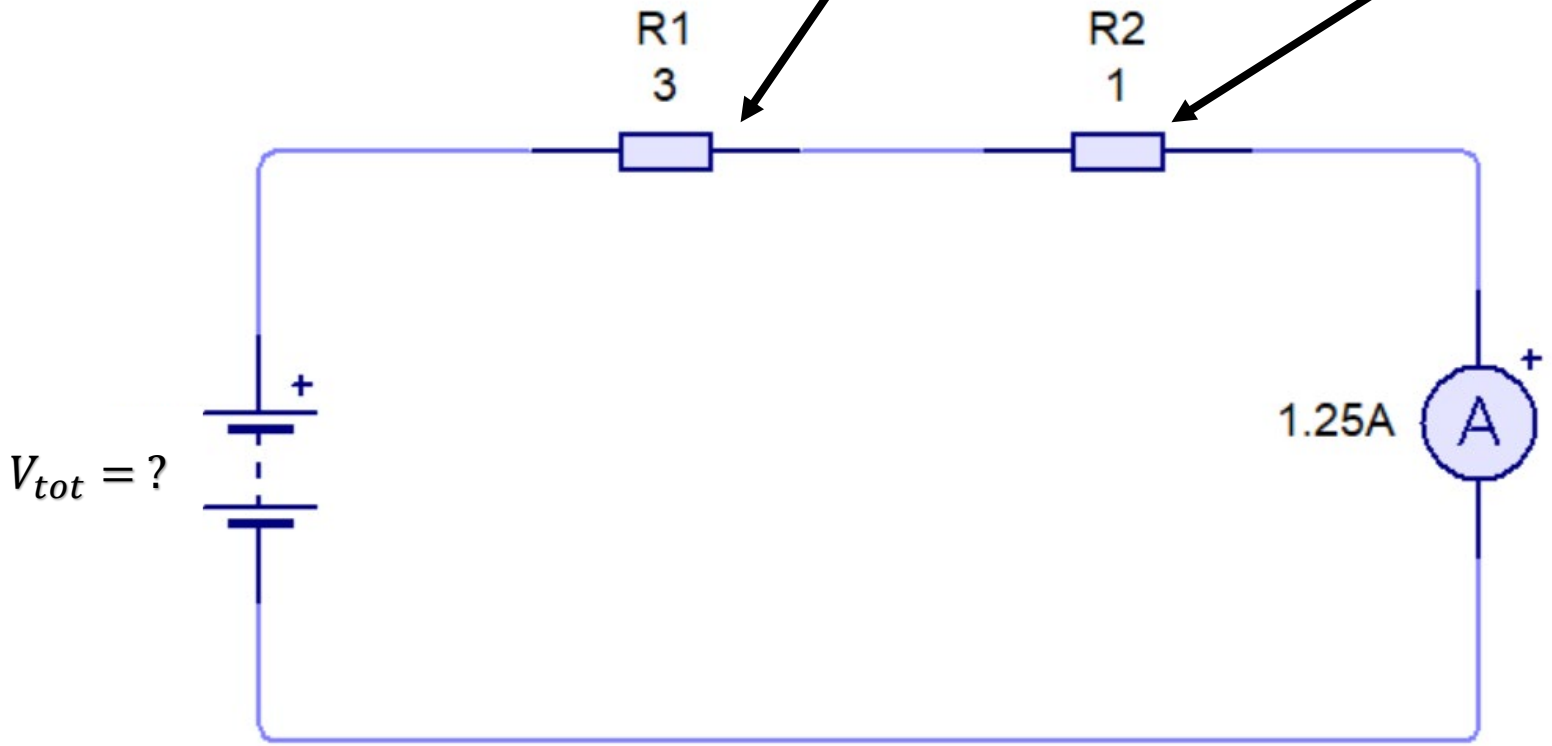


$$5v - 3.59v - 1.41v = 0v$$

KVL Example

$V_1 = 1.25 * 3 = 3.75$

$V_2 = 1.25 * 1 = 1.25$

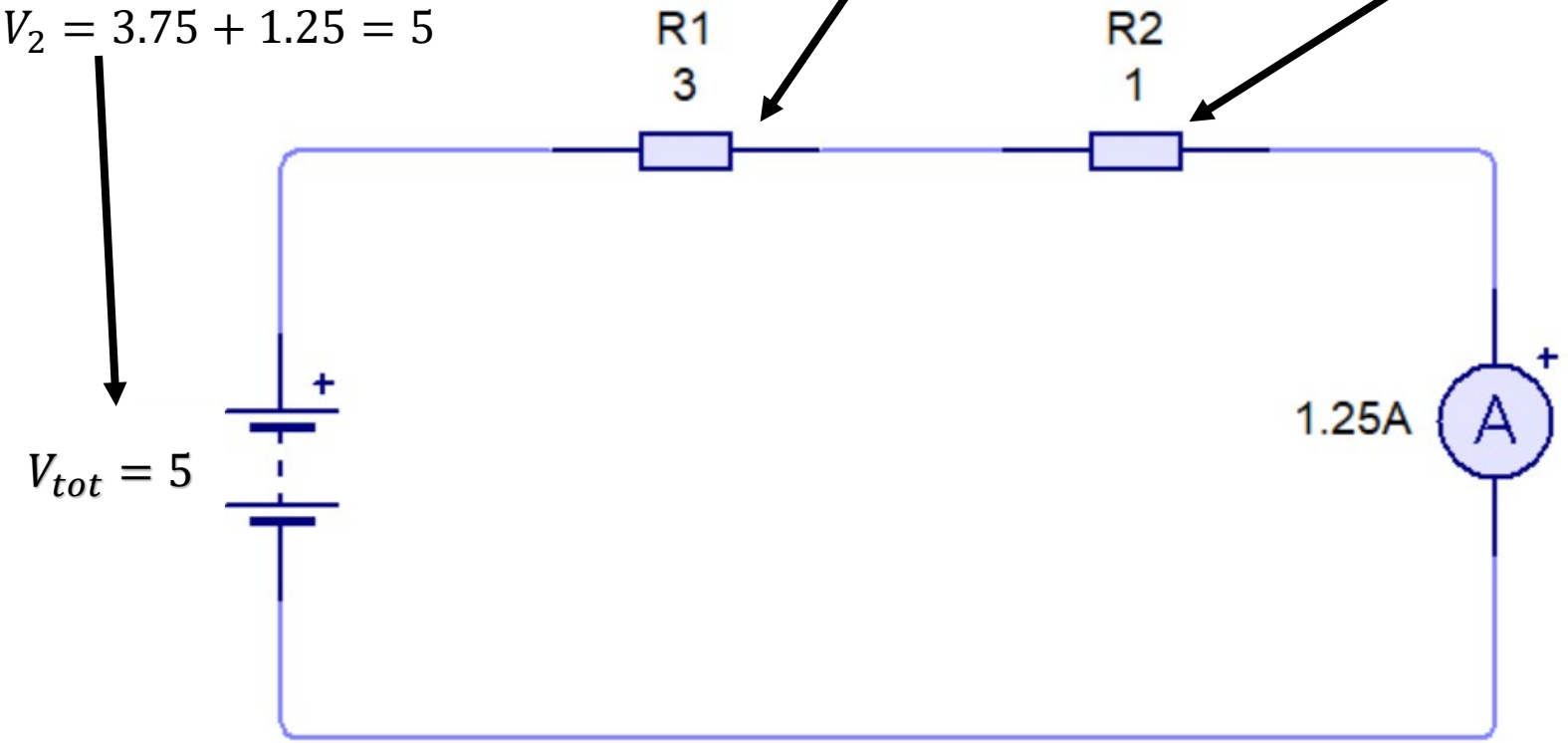


KVL Example

$V_1 = 1.25 * 3 = 3.75$

$V_2 = 1.25 * 1 = 1.25$

$V_{tot} = V_1 + V_2 = 3.75 + 1.25 = 5$



$V_{tot} = 5$